Final Project Expectations – Advanced Optics Fall 2026

The final project replaces a traditional final exam. Teams of two students will analyze or design a complete photonic system. The project must connect at least four course units and include *quantitative* estimates of performance limits. Each team will deliver a short proposal, a conference-style **25 minute** oral presentation, and a **6-to-10-page** conference-style manuscript with references (references not included in page limit).

- Topics should encompass a single application or system but use multiple references for supporting material. The area may be similar but not the same as a Spotlight topic previously presented in the course.
- The project must identify the source, propagation path, modulation or nonlinear element if applicable, detection strategy, dominant noise sources, bandwidth or resolution limit, and at least one realistic engineering tradeoff.
- Both partners or all team members must contribute to both oral and written components and include a short author-contribution statement. You may submit the same final pdf.
- Choose a system for which the team can make quantitative estimates, not merely give a descriptive literature review.
- Include a clear block diagram and explain what happens to the optical field and/or photon flux at each stage.
- Estimate at least three relevant performance quantities, such as signal-to-noise ratio, bandwidth, resolution, optical power budget, nonlinear conversion efficiency, dispersion length, coherence length, detection limit, or link budget.
- Identify the dominant limitation and at least one realistic alternative design choice.
- Use properly cited references and include a short author-contribution statement.

Summary of required components:

1. Introduction and connections to concepts covered in class
2. Identification of optical elements including a block diagram
3. Quantitative estimates of performance quantities that are original work/calculations
4. Identification of dominant limitations or risks and possible alternative design choice
5. Author-contribution statement
6. References

Guiding Grading Rubric

Grade/Skill	C	B	A
Optical System overview	Overview of optical system components	Overview of optical system components provided; some connections made to course material; conceptual understanding of how components interact with optical field	Clear overview of optical system components and connections to course topics; demonstrates accurate technical understanding of how components interact with optical field
System performance evaluation	Quantitative estimate of at least one relevant performance quantity with supporting calculations or numerical simulations	Quantitative estimate of at least one relevant performance quantity with supporting calculations or numerical simulations	Quantitative estimate of at least three relevant performance quantities with supporting calculations or numerical simulations
Introduction	some background from topics covered in class is (briefly) reviewed;	some background from topics covered in class is accurately (briefly) reviewed; demonstrates some understanding of how chosen system fits into course topics	Necessary background from topics covered in class is accurately (briefly) reviewed; demonstrates understanding of how chosen system fits into course topics
Identification of trade-off or limitation	Identification of engineering trade-offs or limitations; qualitative analysis of at least one of them	Identification of engineering trade-offs or limitations; Quantitative analysis of at least one of them	Clear identification of engineering trade-offs or limitations including those not explicitly stated by the original authors; Quantitative analysis of at least one of them
Originality	Original work and ideas presented in quantitative analysis;	Original work and ideas presented in quantitative analysis; demonstrates skills learned in class such as Zemax, Tidy3D or optics-related calculations/simulations	Original work and ideas presented in quantitative analysis are thorough, accurate, and precise; demonstrates skills learned in class such as Zemax, Tidy3D or related calculations/simulations
Format/References	document with references cited	Publication style document with references cited	Publication style document with references cited that are relevant and appropriate
Writing quality	Written communication is difficult to follow or understand logical flow. Ideas and claims are only partially supported	Written communication is satisfactory but possibly disjointed or sometimes unsupported	Written communication is logical and flows from one idea to the next; ideas are supported by evidence